

Planning and action selection

Project: Grounded Learning Service

1. Your mission

Your learning target

I can explain the mechanism and terminology of planning and action selection.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from State machines for bounded agents. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Failure budgets and service objectives. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Testing model-dependent behaviour. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

This week you will practise:

Explain the mechanism and terminology of planning and action selection

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Retrieve what you already know

Recall State machines for bounded agents. Explain why a recorded result needs its input conditions and evidence source before it can support a claim.

My starting explanation

This week's end goal

Explain planning and action selection using the deciding step in this fictional service.

2. Notice the evidence

Planning proposes steps; execution must still validate permissions and prerequisites for each step. Replanning cannot invent authority.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing. These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

List the given facts and one fact that is not supplied.

3. See how it works

For read-only reporting, allowed plan is validate -> retrieve -> verify -> answer. If retrieval lacks evidence, route to review. Reject write or send steps before they run.

Planning and action selection — evidence trace

Input and scope · Check or calculation · Conclusion and limit

Input and scope: Generated plan retrieves a room, books it and then answers. It calls booking necessary for helpfulness.

Check or calculation: For read-only reporting, allowed plan is validate -> retrieve -> verify -> answer. If retrieval lacks evidence, route to review. Reject write or send steps before they run.

Conclusion and limit: Booking is outside the task boundary. Execute only permitted reporting steps, with a fallback if evidence is insufficient.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same inputs. If a required input is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Connect the picture and the explanation

Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

4. Compare the cases

Compare a familiar case with a changed case

Generated plan retrieves a room, books it and then answers. It calls booking necessary for helpfulness. Show the intermediate steps and give the conclusion with its scope.

Keep the procedure fixed. Explain what changed in the inputs and result.

Change one thing in the pictured case

Change: Supply an approved card giving height 24 metres.

Prediction: The bot can retrieve it and answer 24 metres with attribution.

Reason: The missing factual input is now available within the allowed collection.

5. Words that unlock the idea

Word · Meaning

Word: Plan

Meaning: A proposed ordered set of steps

Word: Prerequisite

Meaning: A condition required before a step

Word: Action selection

Meaning: Choosing a permitted operation under the current state

Use two terms to explain a step in this week's example.

6. Follow a worked example

Generated plan retrieves a room, books it and then answers. It calls booking necessary for helpfulness.

Booking is outside the task boundary. Execute only permitted reporting steps, with a fallback if evidence is insufficient.

Compare the intermediate evidence with the final statement. Explain why each number or decision follows from the stated inputs. The conclusion belongs to this scenario and procedure; changing the inputs, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Planning and action selection

Plan around missing evidence. A bot must answer a tower-height question using approved cards only.

The facts we will use

Search finds no height card. Allowed actions: search, clarify, return unsupported. Unapproved web access is excluded.

Planning and action selection

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Name missing input

A supported height answer needs an approved height record.



2. Select allowed action

After the empty search, return unsupported rather than inventing a number.



3. Explain next step

Ask for an approved source card if the question should be answered.

CHANGE ONE THING

Supply an approved card giving height 24 metres.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Name missing input

What happens and why: A supported height answer needs an approved height record.

Step: 2. Select allowed action

What happens and why: After the empty search, return unsupported rather than inventing a number.

Step: 3. Explain next step

What happens and why: Ask for an approved source card if the question should be answered.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Planning selects useful actions within available evidence and authority. More steps cannot create permission to access an excluded source.

What this example does not establish

Source validity still matters; the presence of a card does not independently verify its accuracy.

7. Find and repair a mistake

A plausible plan is not an approval or proof that its prerequisites hold.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself. The first may require a correction; the second requires an explicitly revised method and new evidence. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Locate the earliest unsupported step

A report omits this qualification: A plausible plan is not an approval or proof that its prerequisites hold. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

State one repair and the checks you would rerun.

8. Practise together

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Explain your prediction to a partner before comparing results.

Practise with less help: Trace the worked case

Use workbook W2 for the connected example “Planning and action selection”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case.

Each person writes their own explanation.

Check your reasoning

Use the worked case for Planning and action selection. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Choose a permitted plan when no approved tower-height card exists. What new answer becomes supported after a card records 24 metres, and why could the bot not invent it earlier?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

Predict, investigate and preserve the evidence

Generated plan retrieves a room, books it and then answers. It calls booking necessary for helpfulness. Show the intermediate steps and give the conclusion with its scope.

Record · Your evidence

Record: Given input and fixed rule

Your evidence: _____

Record: Prediction before checking

Your evidence: _____

Record: Actual result, or “not run”

Your evidence: _____

Record: First mismatch and one explanation

Your evidence: _____

Record: Boundary or missing input and expected response

Your evidence: _____

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

Explain → apply → challenge

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

A report omits this qualification: A plausible plan is not an approval or proof that its prerequisites hold. Explain how omitting it could change a reader’s interpretation, and write an accurate replacement.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

Explain which evidence supports your conclusion and which claim remains unproved.

11. Connect your project

Add a labelled planning and action selection evidence sheet to Grounded Learning Service. Show how you can explain the mechanism and terminology of planning and action selection. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence · Record

Project evidence: Component and version

Record: _____

Project evidence: Inputs permitted at prediction time

Record: _____

Project evidence: Trace and comparison with earlier work

Record: _____

Project evidence: Failure, limit and human review

Record: _____

Evidence for your project

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or evidence that makes the deciding step valid, so another team can review it.

12. Show what you know

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer. For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

My limited claim and a useful next check

13. Study a complete case

Complete evidence record

Planning and action selection — evidence trace

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Conclusion and limit: Booking is outside the task boundary. Execute only permitted reporting steps, with a fallback if evidence is insufficient.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the input through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

14. Try a new case independently

Independent case: Apply a new case independently

Plan validates input then sends a message before checking recipient approval. The task authorises drafting only. Locate violation.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

Plan validates input then sends a message before checking recipient approval. The task authorises drafting only. Locate violation.

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.